

Week 4: Work and Kinetic Energy – Tutorial Sheet

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Use the Master Procedure: identify, mark the two states, draw the free-body diagram, find the work of each force with its sign, predict, select the method, execute $W_{\text{tot}} = \Delta K$, evaluate. Use $g = 9.80 \text{ m/s}^2$. Problems A1 and A2 are solved in full; final answers to the others are at the end.

A. Worked examples

A1. A sled on level ground

A 15.0 kg sled is pulled 10.0 m along level ground by a rope with tension 60 N, 20° above the horizontal. The coefficient of kinetic friction is 0.10. Find the work done by each force, the total work, and the final speed if it starts from rest.

Solution. Vertical: $n = mg - T \sin 20^\circ = 147 - 20.5 = 126.5 \text{ N}$, so $f_k = 0.10(126.5) = 12.65 \text{ N}$. $W_T = Ts \cos 20^\circ = (60)(10.0)(0.940) = +564 \text{ J}$; $W_f = -(12.65)(10.0) = -127 \text{ J}$; $W_w = W_n = 0$. Total: 437 J. Check (Method II): $(60 \cos 20^\circ - 12.65)(10.0) = (56.4 - 12.65)(10.0) = 437 \text{ J}$. Work-energy: $437 = \frac{1}{2}(15.0)v^2 \Rightarrow v = 7.6 \text{ m/s}$.

A2. A spring-launched ball

A 0.300 kg ball is pushed against a horizontal spring ($k = 500 \text{ N/m}$), compressing it 0.20 m, and released on a smooth surface. Find its speed when it leaves the spring.

Solution. Work done by the spring: $\frac{1}{2}kx^2 = \frac{1}{2}(500)(0.20)^2 = 10.0 \text{ J}$. This is the total work (no friction), so $\frac{1}{2}mv^2 = 10.0$ and $v = \sqrt{2(10.0)/0.300} = 8.2 \text{ m/s}$.

B. Practice

B1. A ramp with friction

A 10.0 kg crate slides 4.0 m down a 30° ramp with $\mu_k = 0.25$, starting from rest. Find the work done by gravity, by friction and by the normal force, and the final speed.

B2. Stopping a truck

A 8000 kg truck at 20 m/s is stopped by a constant braking force of 32 kN. Find the stopping distance. How far would it need at 40 m/s?

B3. A varying force

The force on a particle is $F_x = 10x - 2x^2$ (SI units). Find the work done from $x = 0$ to $x = 3.0 \text{ m}$.

B4. A spring and friction

A 1.0 kg block is pressed against a spring ($k = 200 \text{ N/m}$) compressed 0.30 m and released on a rough floor ($\mu_k = 0.30$). Find its speed as it leaves the spring and the total distance it slides.

B5. Power of a lift

A motor raises an 800 kg load at a constant 2.0 m/s. What is the power developed (neglecting friction)? How much energy does it supply in one minute?

B6. Challenge: the method is not given

A 60 kg skier slides down a 25 m long slope inclined 20° to the horizontal with $\mu_k = 0.10$, and then continues along a level patch with $\mu_k = 0.20$ until she stops. How far does she slide on the level patch?

Answers

B1 $W_g = +196 \text{ J}$, $W_f = -85 \text{ J}$, $W_n = 0$; $v = 4.7 \text{ m/s}$

B2 50 m; 200 m (four times)

B3 27 J

B4 $v = 4.0 \text{ m/s}$; $d = 3.1 \text{ m}$

B5 $P = mgv = 15.7 \text{ kW}$ ($\approx 21 \text{ hp}$); $9.4 \times 10^5 \text{ J}$

B6 kinetic energy at the bottom $3.6 \times 10^3 \text{ J}$ ($v = 11 \text{ m/s}$); distance on the level patch 31 m